

# Astro-3\_M\_down CC



# Astro-3\_M\_down BP

myelin assembly

ensheathment of neurons

myelin assembly

axon ensheathment

myelination

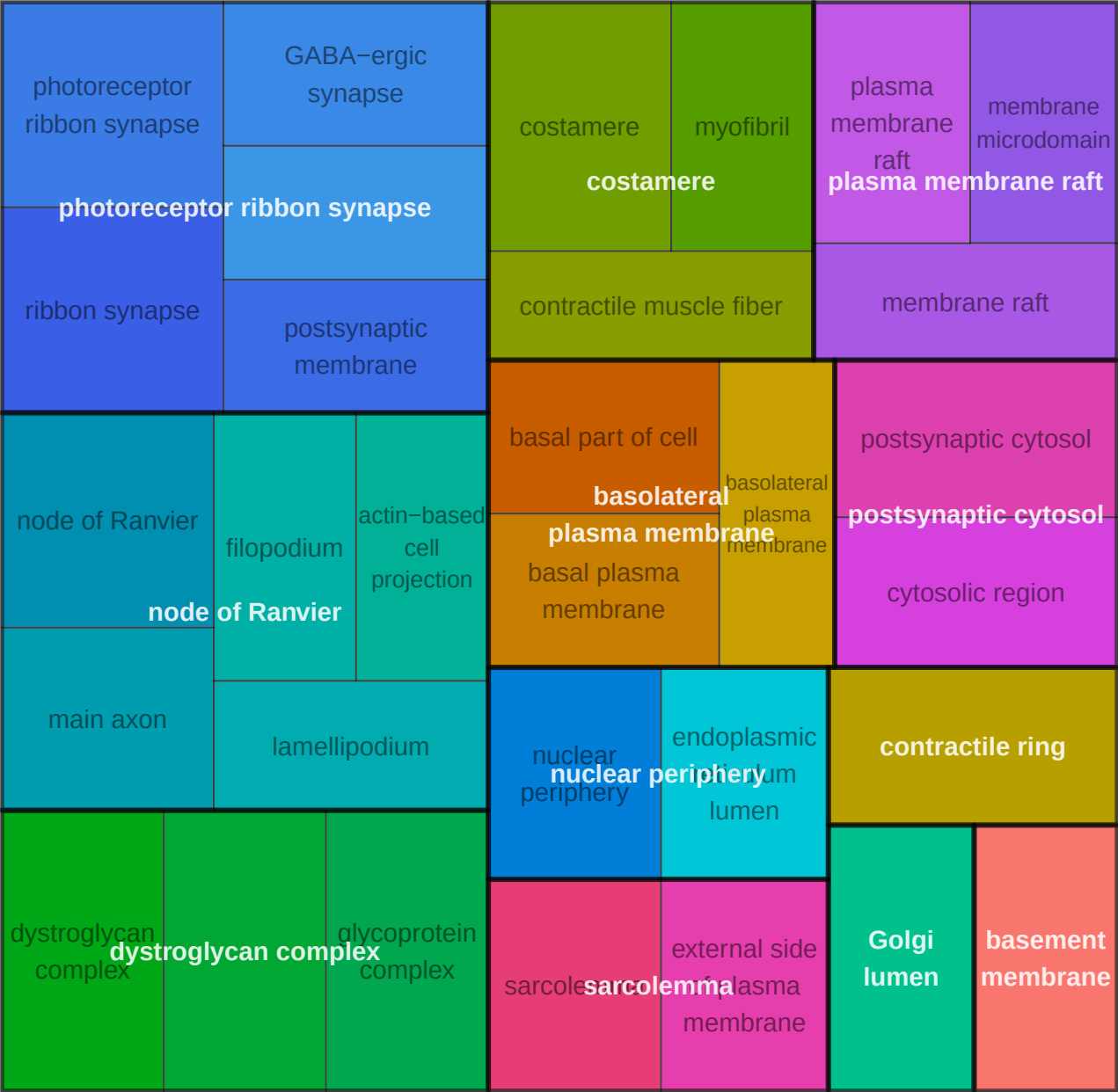
positive  
regulation  
of bone  
resorption

regulation of endocytosis

regulation of endocytosis

regulation of vesicle-mediated transport

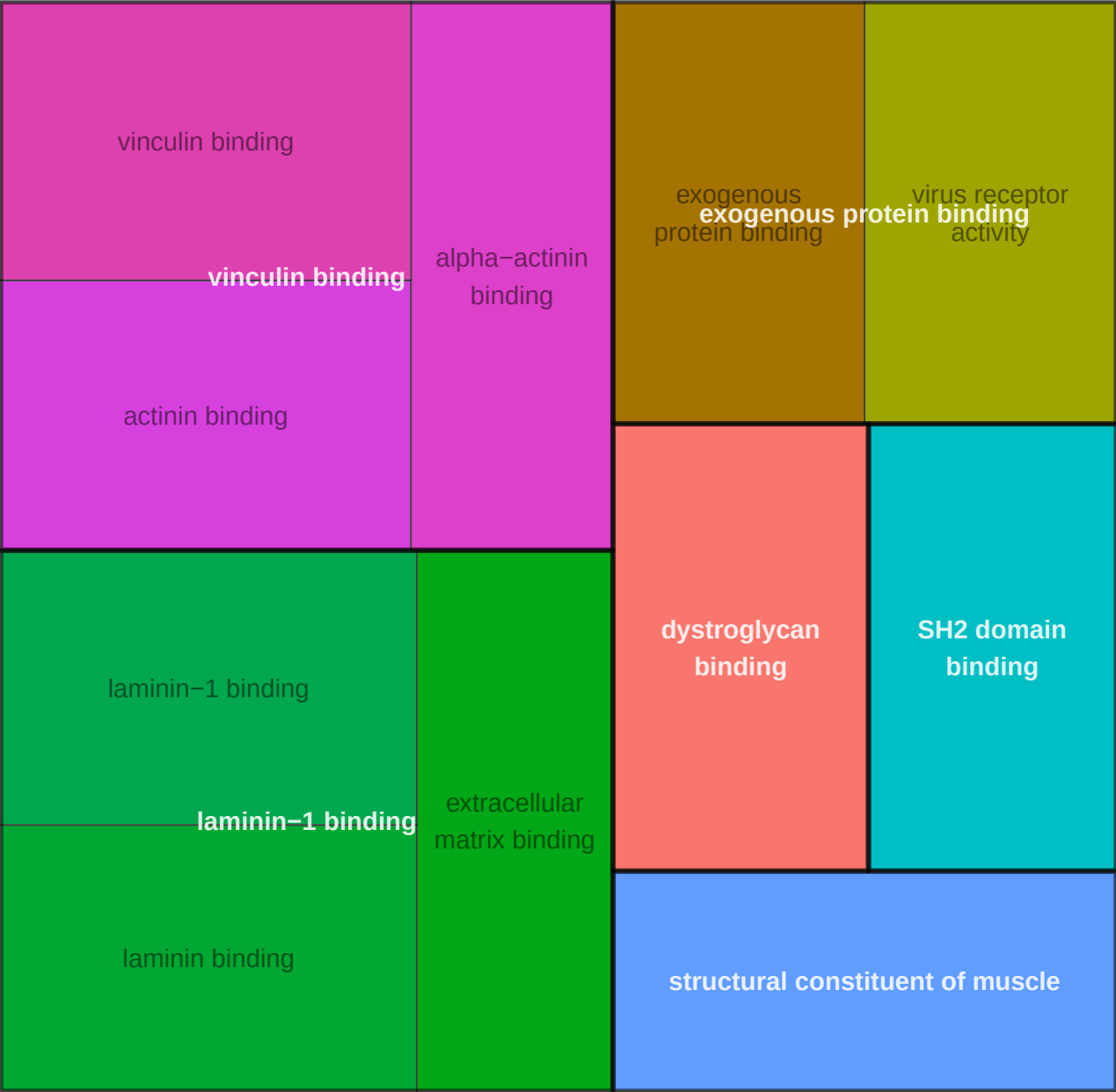
Astro-3\_M\_up CC



# Astro-3\_M\_up BP



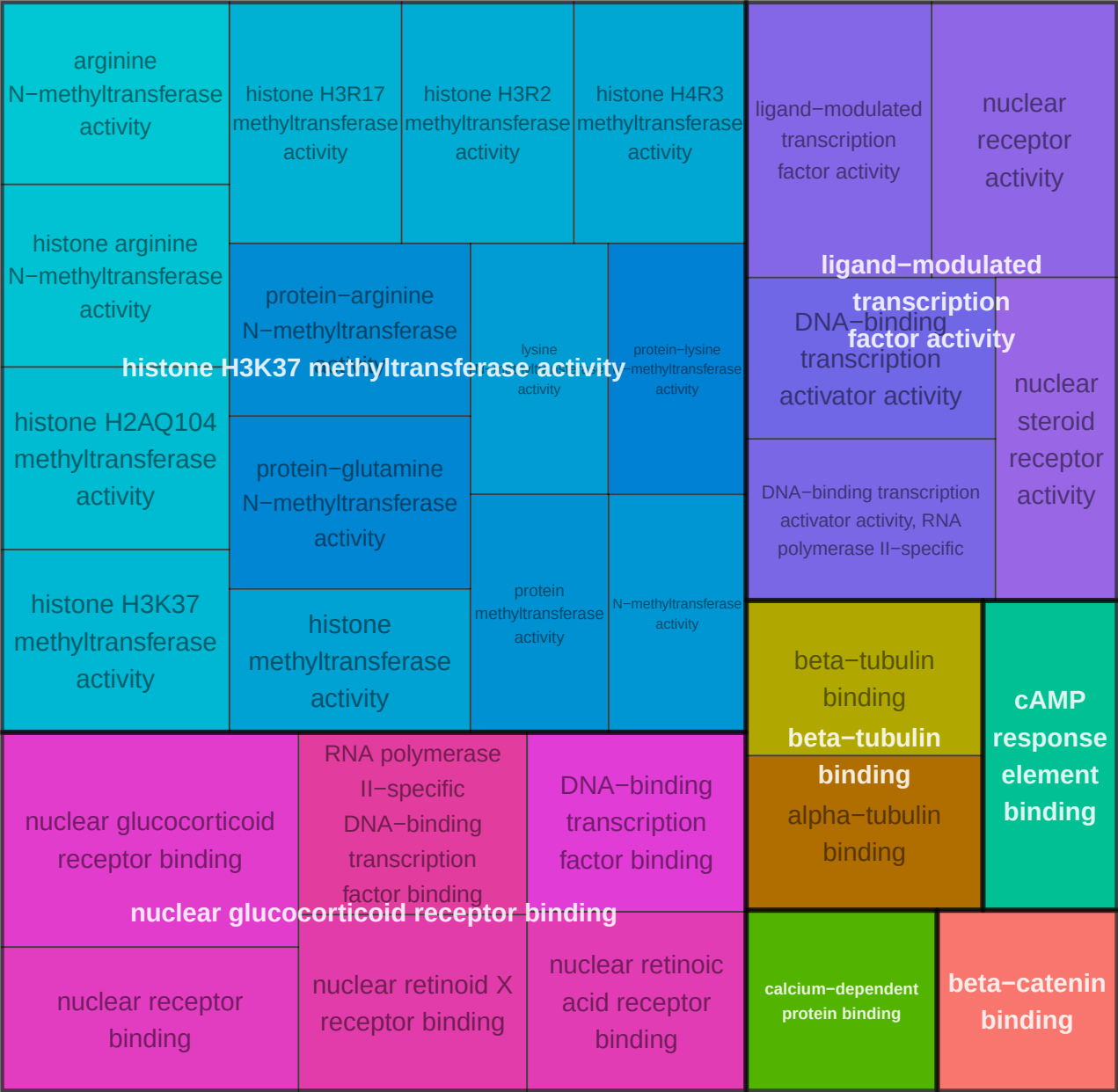
Astro-3\_M\_up MF



Astro-4\_F\_up BP



Astro-4\_F\_up MF



Excit-L2\_5-1\_0\_downF\_DownM BP

|                                                    |                                                                |                                                              |                                                       |                                                       |                                                                      |
|----------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------|
| endosome to plasma membrane protein transport      | establishment of protein localization to postsynaptic membrane | neurotransmitter receptor transport to postsynaptic membrane | positive regulation of neuron projection arborization | neuron projection arborization                        | dendritic spine development                                          |
| protein localization to postsynapse                | endosome to plasma membrane protein transport                  | establishment of protein localization to plasma membrane     | regulation of neuron projection arborization          | positive regulation of neuron projection arborization | dendritic spine organization                                         |
|                                                    |                                                                |                                                              |                                                       |                                                       | neuron projection organization                                       |
| protein localization to postsynaptic membrane      | protein localization to synapse                                | protein localization to cell junction                        | acetyl-CoA biosynthetic process                       | acetyl-CoA metabolic process                          | neurotransmitter receptor transport to plasma membrane               |
| positive regulation of peptidyl-lysine acetylation | positive regulation of peptidyl-lysine acetylation             | regulation of protein acetylation                            | acyl-CoA biosynthetic process                         | thioester biosynthetic process                        | neurotransmitter receptor transport, endosome to plasma membrane     |
|                                                    |                                                                |                                                              |                                                       |                                                       | neurotransmitter receptor transport, endosome to plasma membrane     |
|                                                    |                                                                |                                                              |                                                       |                                                       | receptor localization to synapse                                     |
| positive regulation of peptidyl-lysine acetylation | positive regulation of peptidyl-lysine acetylation             | protein acetylation                                          | acyl-CoA metabolic process                            | thioester metabolic process                           | regulation of postsynaptic membrane neurotransmitter receptor levels |
| regulation of peptidyl-lysine acetylation          | protein acetylation                                            | protein acylation                                            | neuropeptide signaling pathway                        |                                                       |                                                                      |



# Excit-L2\_5-1\_0\_downF\_DownM MF

acid-thiol ligase activity

ligase activity, forming  
carbon-sulfur bonds

CoA-ligase activity

CoA-ligase activity

ligase activity

beta-catenin  
binding

signaling receptor  
complex adaptor activity

signaling receptor complex adaptor activity

# Excit-L2\_5-1\_0\_upF\_DownM CC

GINS complex

GINS complex

DNA replication preinitiation complex

protein-DNA  
complex

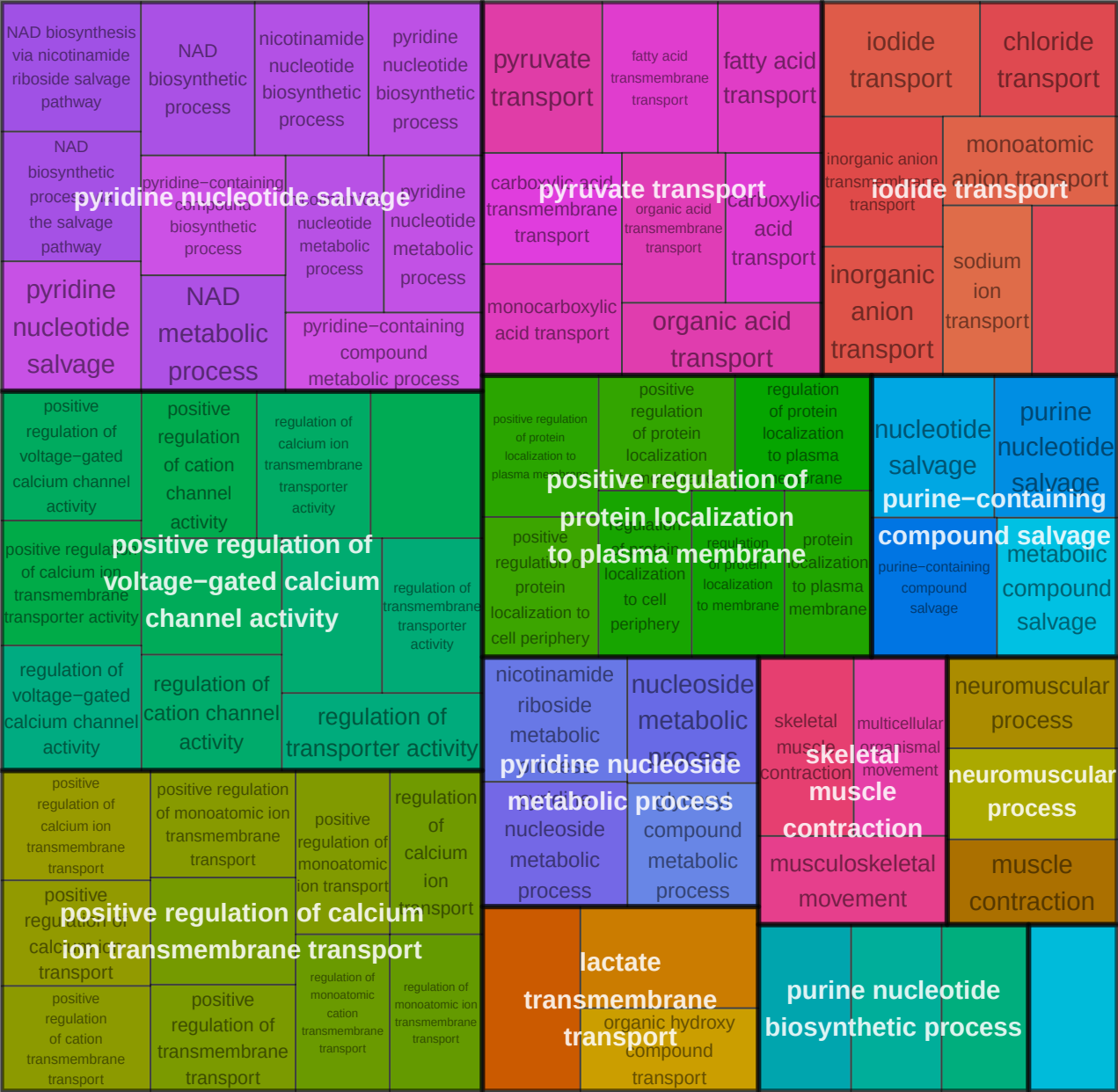
nuclear chromosome

sarcolemma

cytoplasmic side of  
cytoplasmic side of membrane

cytoplasmic side of  
plasma membrane

Excit-L2\_5-1\_0\_upF\_DownM BP



Excit-L2\_5-1\_0\_upF\_DownM MF

|                                               |                                                                     |                                                     |                                                  |                                                     |                                                        |
|-----------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| iodide transmembrane transporter activity     | secondary active monocarboxylate transmembrane transporter activity |                                                     | organic acid:sodium symporter activity           |                                                     | monocarboxylic acid transmembrane transporter activity |
|                                               | lactate transmembrane transporter activity                          | solute:sodium symporter activity                    | monoatomic anion channel activity                | solute:monoatomic cation symporter activity         | carboxylic acid transmembrane transporter activity     |
| iodide transmembrane transporter activity     |                                                                     |                                                     |                                                  |                                                     |                                                        |
| fatty acid transmembrane transporter activity | inorganic anion transmembrane transporter activity                  | monoatomic anion transmembrane transporter activity | sodium ion transmembrane transporter activity    | symporter activity                                  |                                                        |
| monocarboxylate:sodium symporter activity     | lipid transporter activity                                          | organic acid transmembrane transporter activity     | organic anion transmembrane transporter activity | secondary active transmembrane transporter activity |                                                        |

## Excit-L2\_5-1\_M\_up CC

specific granule membrane

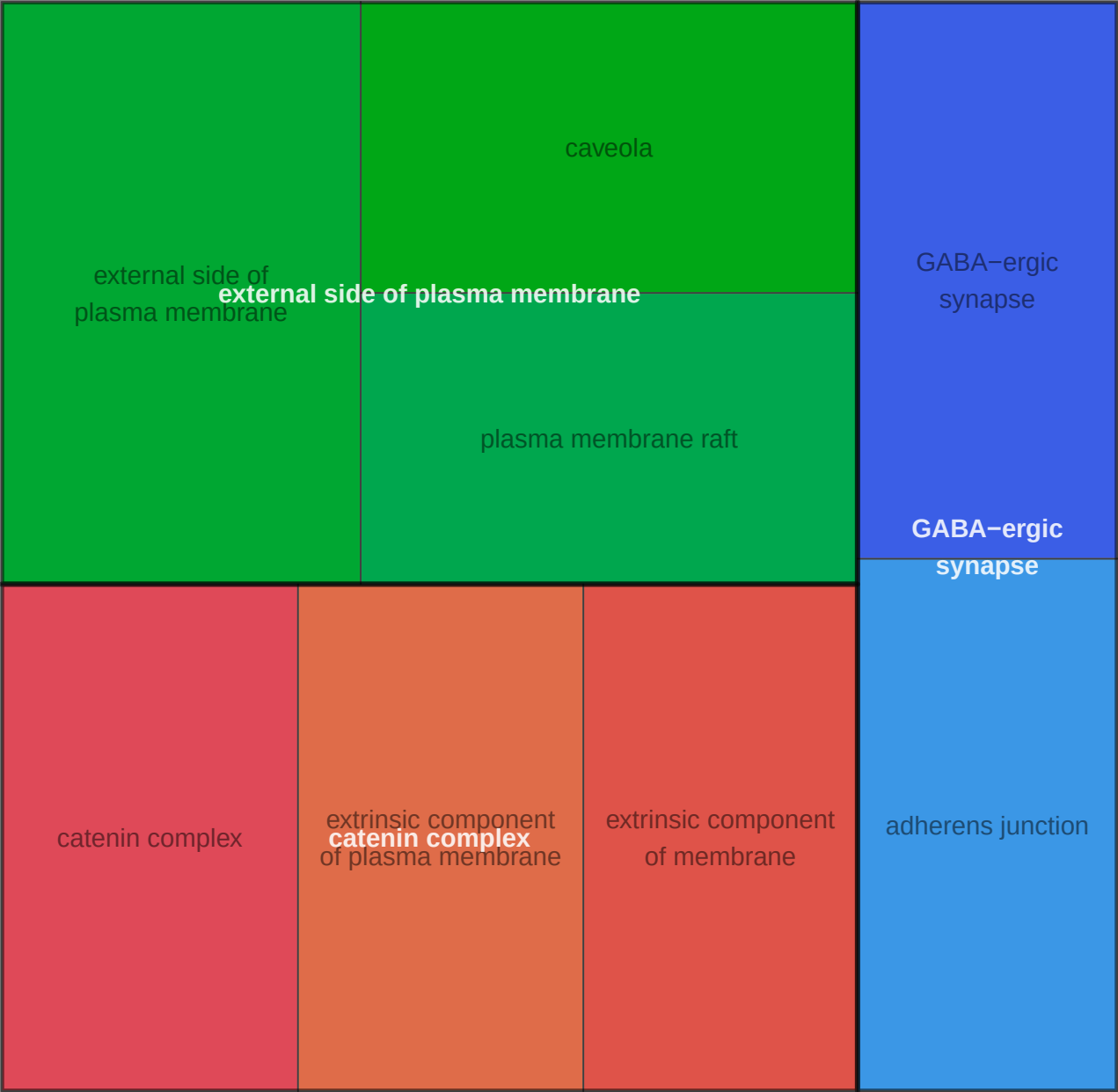
**specific granule membrane**

secondary granule membrane

# Excit-L2\_5-1\_M\_up MF



Excit-L2\_5-2\_M\_down CC



Excit-L2\_5-2\_M\_down BP





|                                                                                   |  |                                                                          |  |                                                      |  |                                  |  |                                                   |  |                                                                                                 |  |                                                                           |  |
|-----------------------------------------------------------------------------------|--|--------------------------------------------------------------------------|--|------------------------------------------------------|--|----------------------------------|--|---------------------------------------------------|--|-------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------|--|
| enteroendocrine cell differentiation                                              |  | pancreatic A cell differentiation                                        |  | type B pancreatic cell differentiation               |  | cold-induced thermogenesis       |  | positive regulation of cold-induced thermogenesis |  | negative regulation of transmembrane receptor protein serine/threonine kinase signaling pathway |  | Notch signaling pathway                                                   |  |
| neuroendocrine differentiation                                                    |  | pancreatic A cell differentiation                                        |  | columnar/cuboidal epithelial cell differentiation    |  | epithelial cell development      |  | positive regulation of cold-induced thermogenesis |  | negative regulation of transmembrane receptor protein serine/threonine kinase signaling pathway |  | Notch signaling pathway                                                   |  |
| neuroepithelial cell differentiation                                              |  | endoderm development                                                     |  |                                                      |  |                                  |  | regulation of cold-induced thermogenesis          |  | adaptive thermogenesis                                                                          |  | adenylate cyclase-activating G protein-coupled receptor signaling pathway |  |
| negative regulation of transforming growth factor beta receptor signaling pathway |  | regulation of transforming growth factor beta receptor signaling pathway |  | cellular response to transforming growth factor beta |  | endocrine system development     |  | hepaticobiliary system development                |  | regulation of cell-matrix adhesion                                                              |  | cell-matrix adhesion                                                      |  |
| regulation of cellular response to transforming growth factor beta stimulus       |  | regulation of growth factor beta receptor signaling pathway              |  | cellular response to transforming growth factor beta |  | liver development                |  |                                                   |  | regulation of cell-substrate adhesion                                                           |  |                                                                           |  |
| spleen development                                                                |  | hematopoietic or lymphoid organ development                              |  | B cell activation                                    |  | glucose metabolic process        |  | hexose metabolic process                          |  | endocrine pancreas development                                                                  |  | temperature homeostasis                                                   |  |
| B cell differentiation                                                            |  | immune system development                                                |  |                                                      |  | monosaccharide metabolic process |  |                                                   |  | endocrine pancreas development                                                                  |  | cell fate commitment                                                      |  |

# Excit-L5-1\_0\_downF\_UpM MF

hormone binding

**hormone binding**

peptide hormone binding

amide binding

G protein-coupled peptide  
receptor activity

**peptide receptor activity**

peptide receptor activity

# Excit-L5-2\_0\_downF\_UpM CC



Excit-L5-2\_0\_downF\_UpM BP



Excit-L5-2\_0\_downF\_UpM MF



# Excit-L5-2\_M\_down CC

germ cell nucleus

PML body

germ cell nucleus

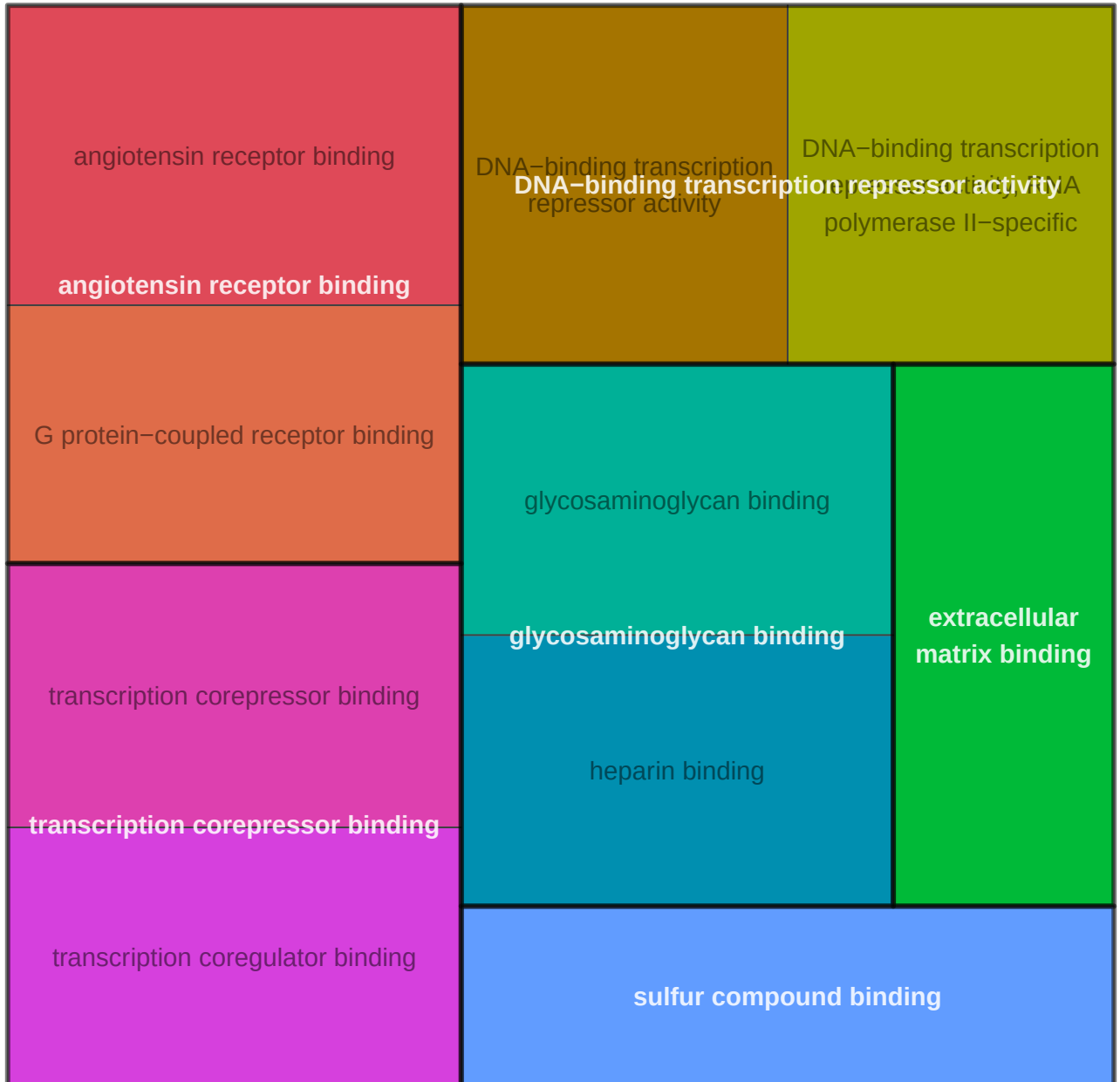
male germ cell nucleus

transcription repressor complex

basement membrane



# Excit-L5-2\_M\_down MF





Excit-L5-2\_M\_up CC

adherens junction

**adherens junction**

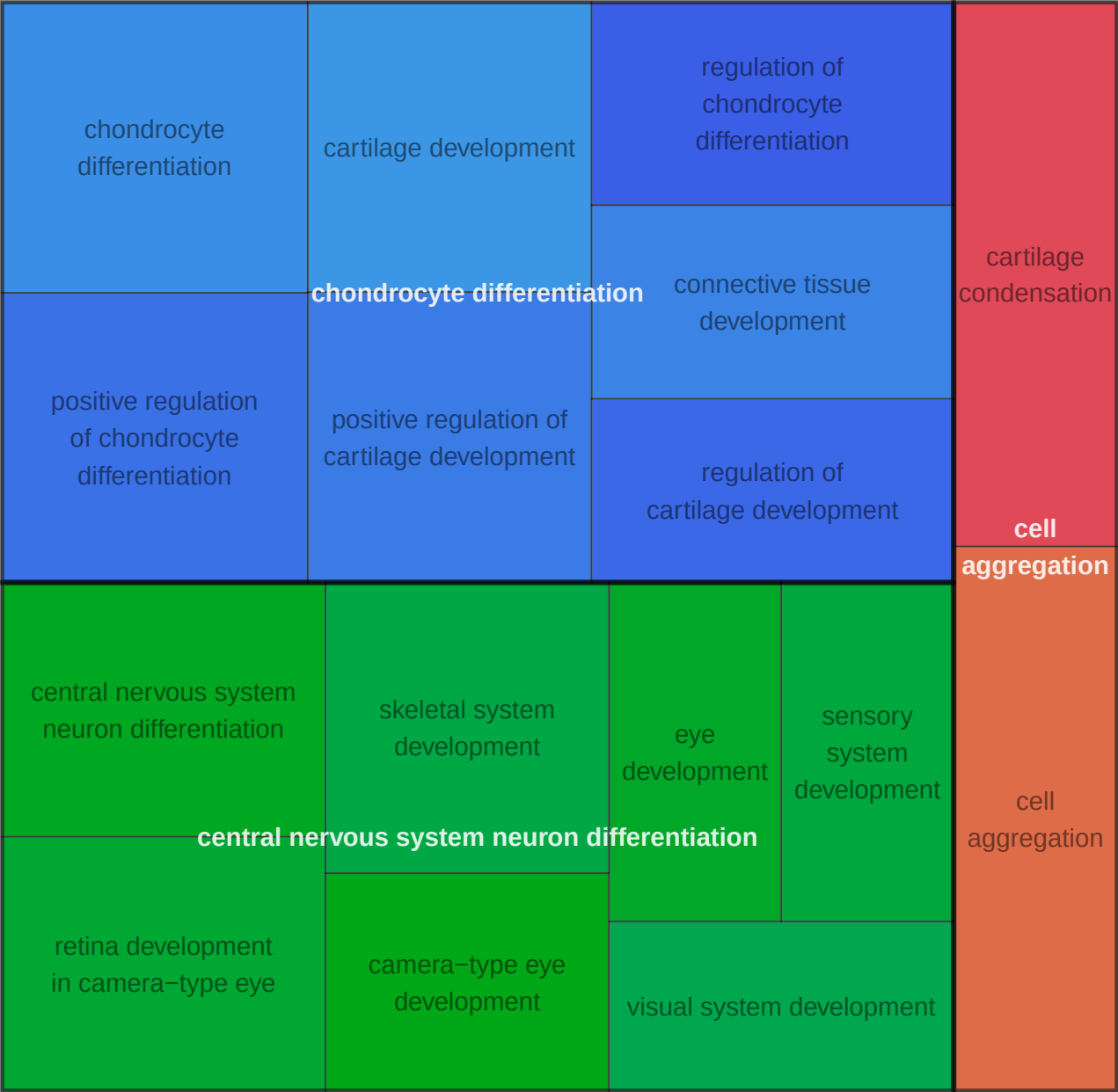
apical junction complex

cell-cell junction

**HFE-transferrin  
receptor  
complex**

**cerebellar mossy fiber**

Excit-L5-2\_M\_up BP



# Macro\_M\_down CC

cell-substrate junction

semaphorin receptor complex

cell-substrate junction

focal adhesion

sorting endosome

neurofilament

cell body membrane

# Macro\_M\_down BP



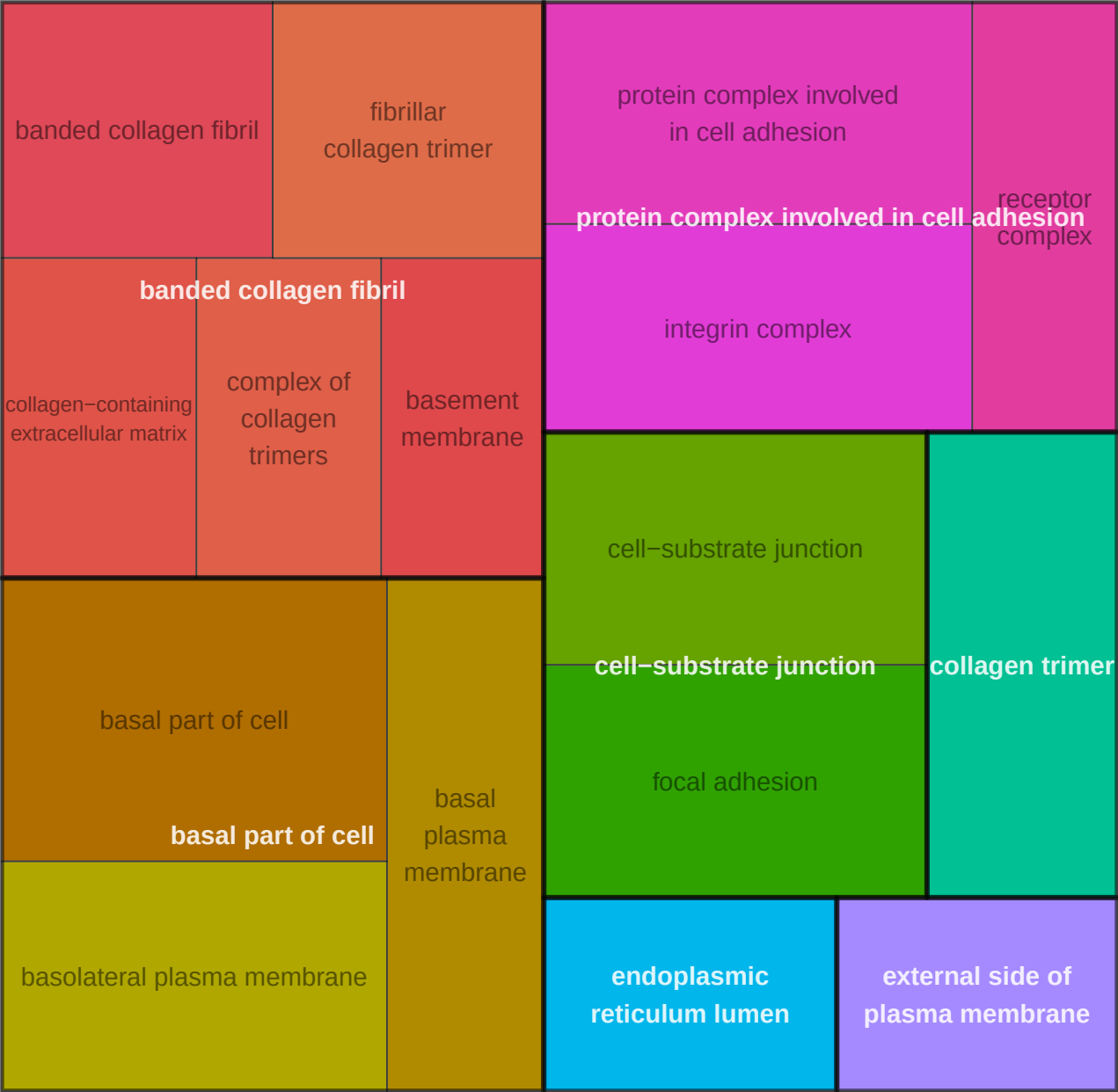
Macro\_M\_down MF



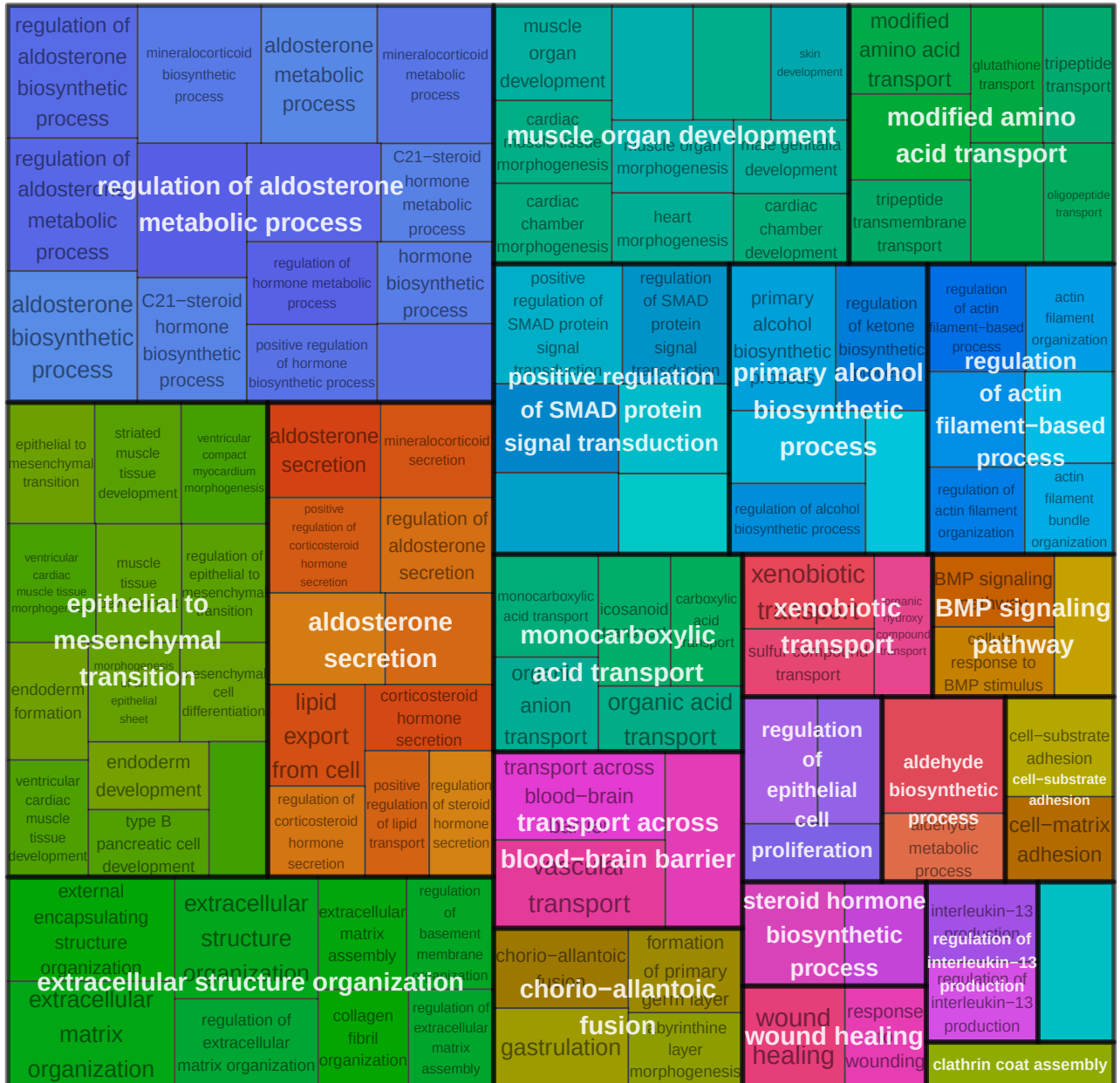
Macro\_M\_up MF



Oligo-3\_0\_downF\_UpM CC

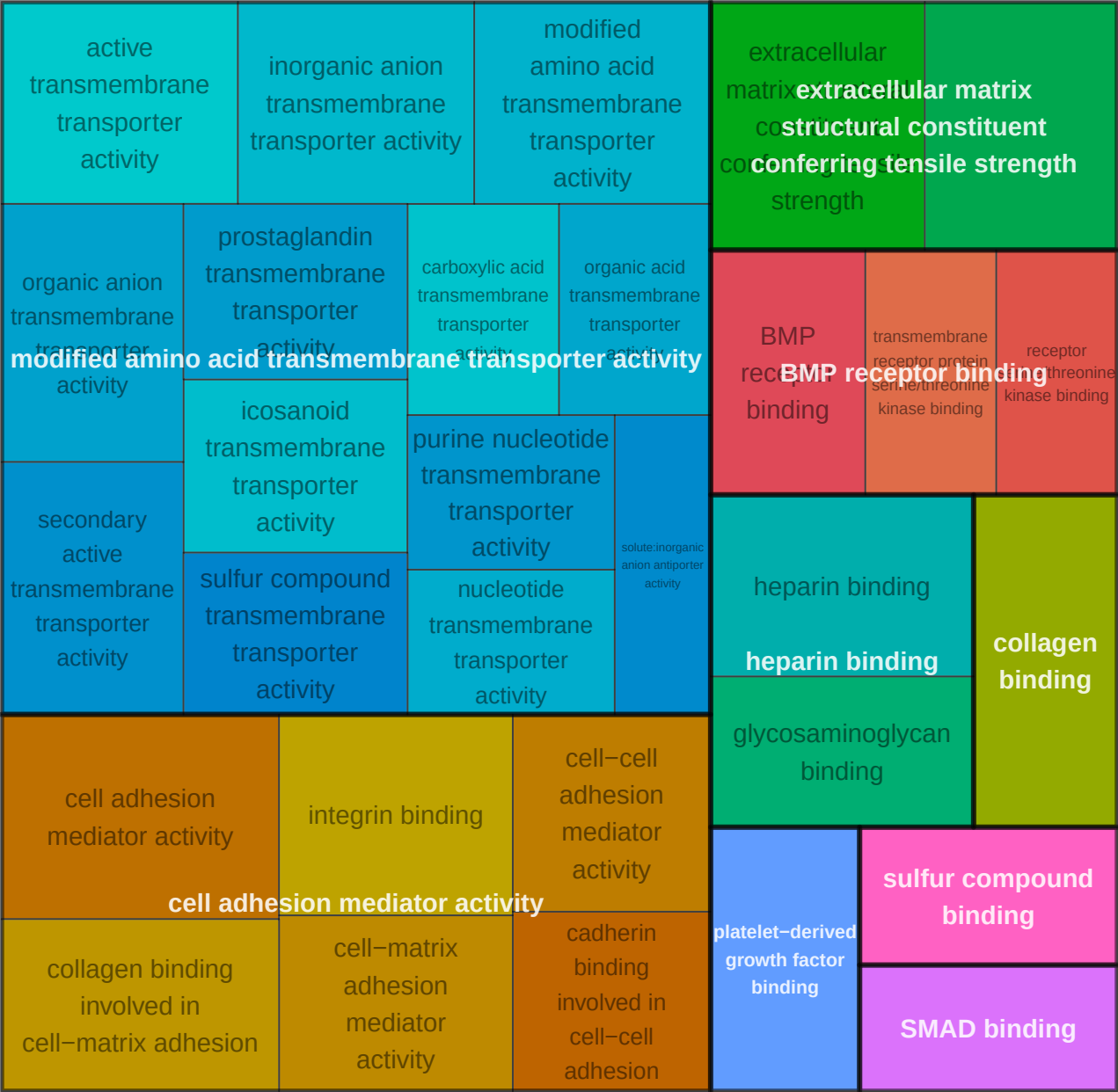


Oligo-3\_0\_downF\_UpM BP





Oligo-3\_0\_downF\_UpM MF



Oligo-3\_0\_upF\_DownM CC

collagen type IX trimer **FACIT collagen trimer** FACIT collagen trimer

Oligo-3\_0\_upF\_DownM BP

deoxyribonucleoside  
triphosphate  
catabolic process

purine nucleoside  
triphosphate  
catabolic process

deadenylation-independent  
decapping of  
nuclear-transcribed mRNA

purine nucleoside triphosphate catabolic process

**nuclear-transcribed  
mRNA catabolic  
process,  
deadenylation-independent  
decay**

deoxyribonucleoside  
triphosphate  
metabolic process

nucleoside triphosphate  
catabolic process

nuclear-transcribed  
mRNA catabolic  
process,  
deadenylation-independent  
decay

# Oligo-3\_0\_upF\_DownM MF

cis-trans isomerase activity

**cis-trans isomerase activity**

peptidyl-prolyl cis-trans isomerase activity

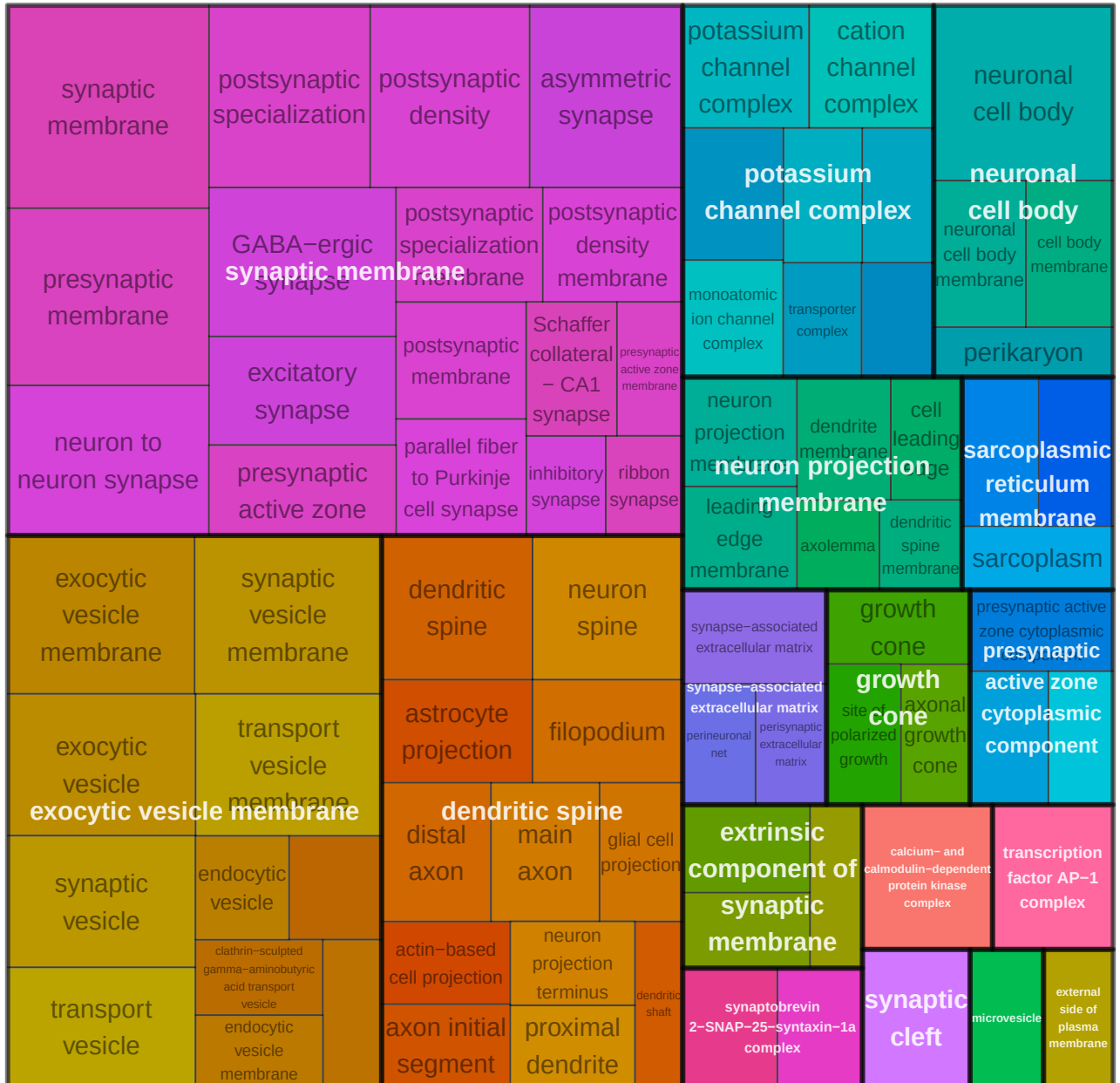
**cyclosporin A binding**

**nucleoside triphosphate  
diphosphatase activity**

**extracellular matrix structural  
constituent conferring tensile strength**

**nucleosome binding**

Oligo-3\_0\_upF\_UpM CC



Oligo-3\_0\_upF\_UpM BP



## Oligo-3\_0\_upF\_UpM MF

voltage-gated monoatomic ion  
channel activity involved in  
regulation of presynaptic  
membrane potential

**voltage-gated monoatomic ion channel activity involved  
in regulation of presynaptic membrane potential**

metal ion transmembrane  
transporter activity

**glutamate receptor binding**

monoatomic cation channel activity

**calcium-dependent phospholipid binding**

**SNARE binding**

# Oligo-3\_M\_down CC

myelin sheath **myelin sheath**

compact myelin

neuromuscular junction

**neuromuscular junction**

cell-cell junction

main axon **main axon**

node of Ranvier

**cell cortex**



## Oligo-3\_M\_down BP

axon ensheathment

myelination

axon  
ensheathment  
in central  
nervous system

axon ensheathment

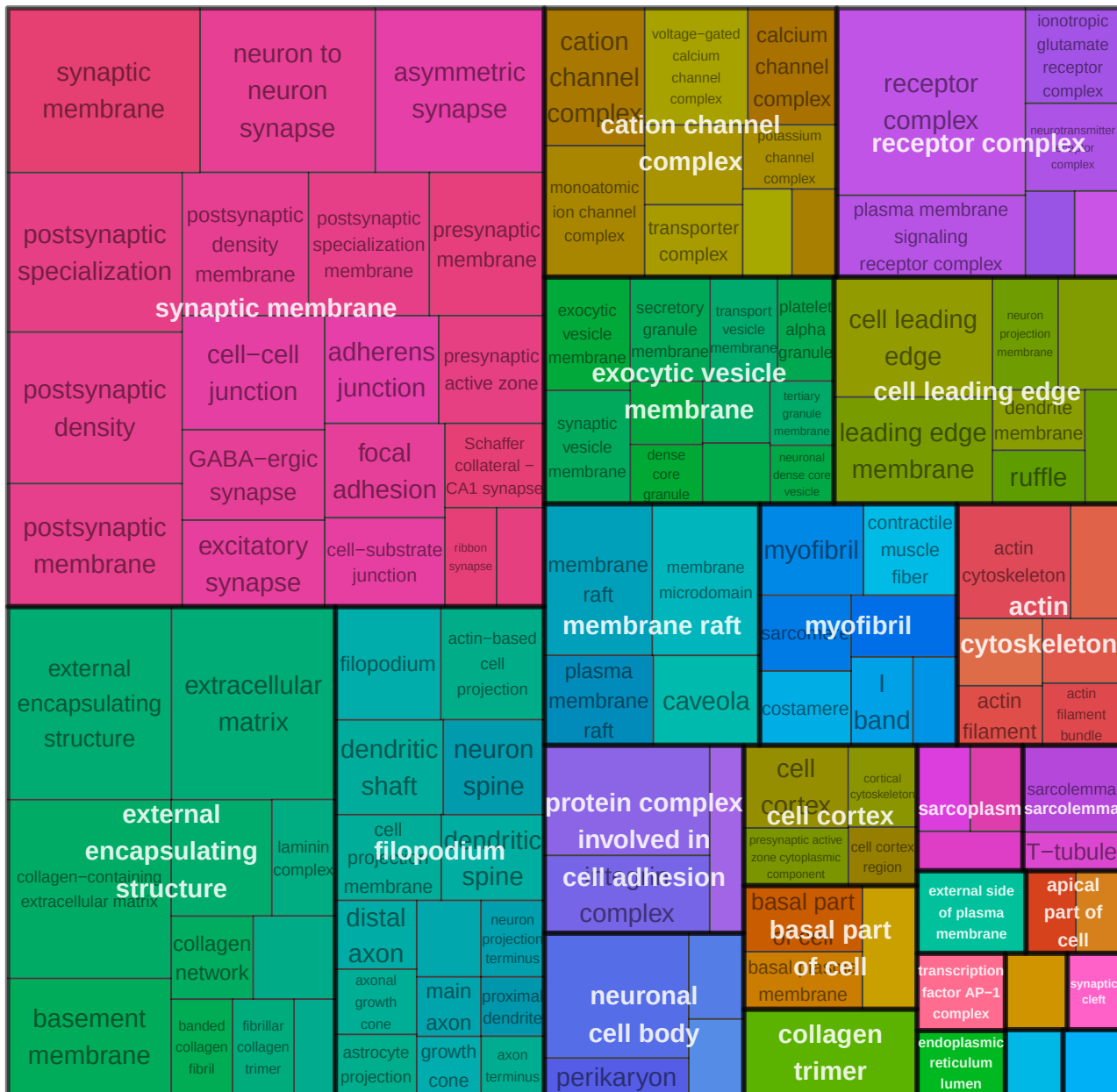
central nervous  
system myelination

ensheathment of neurons

oligodendrocyte  
development

myelin assembly

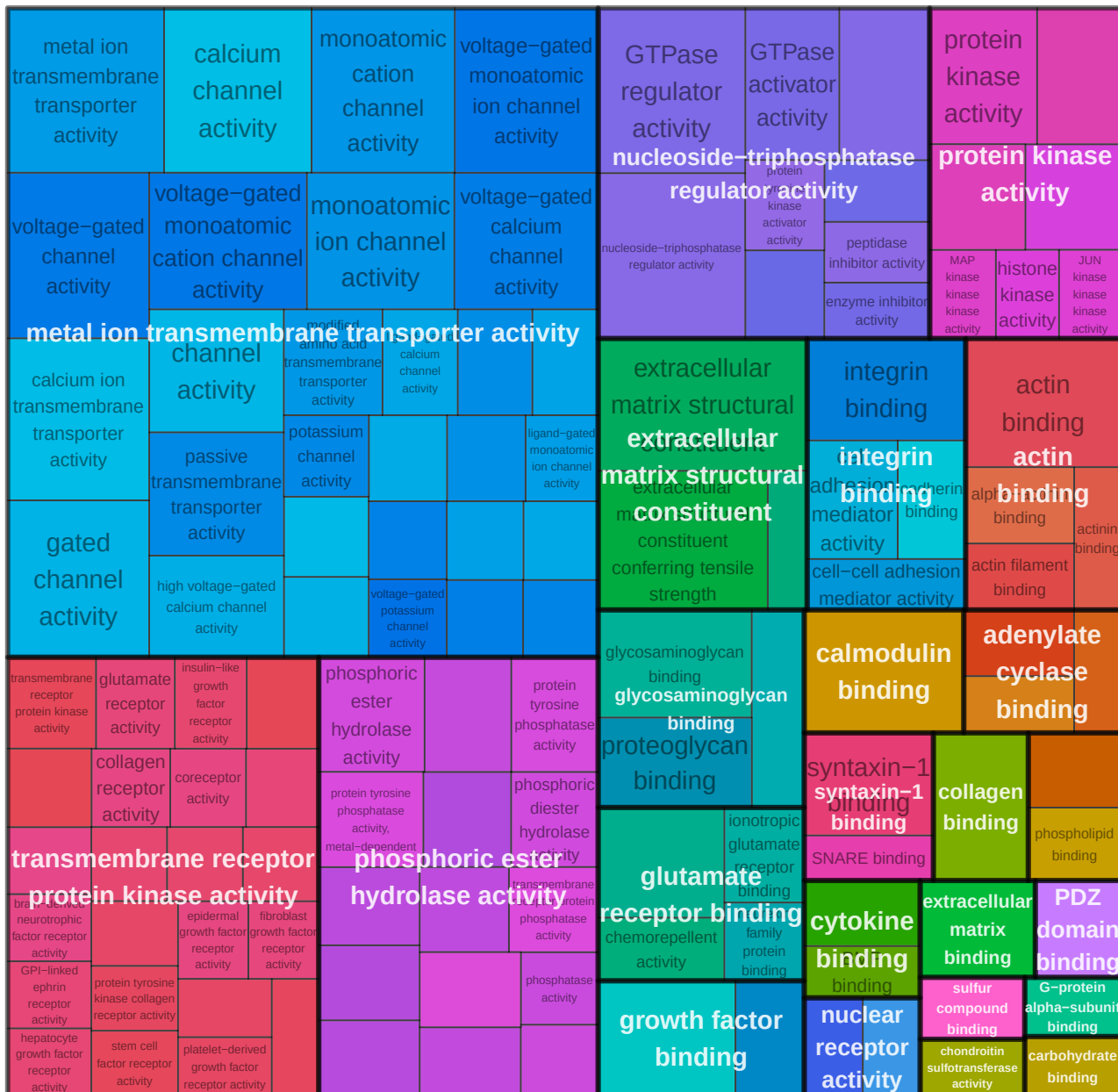
Oligo-3 M up CC



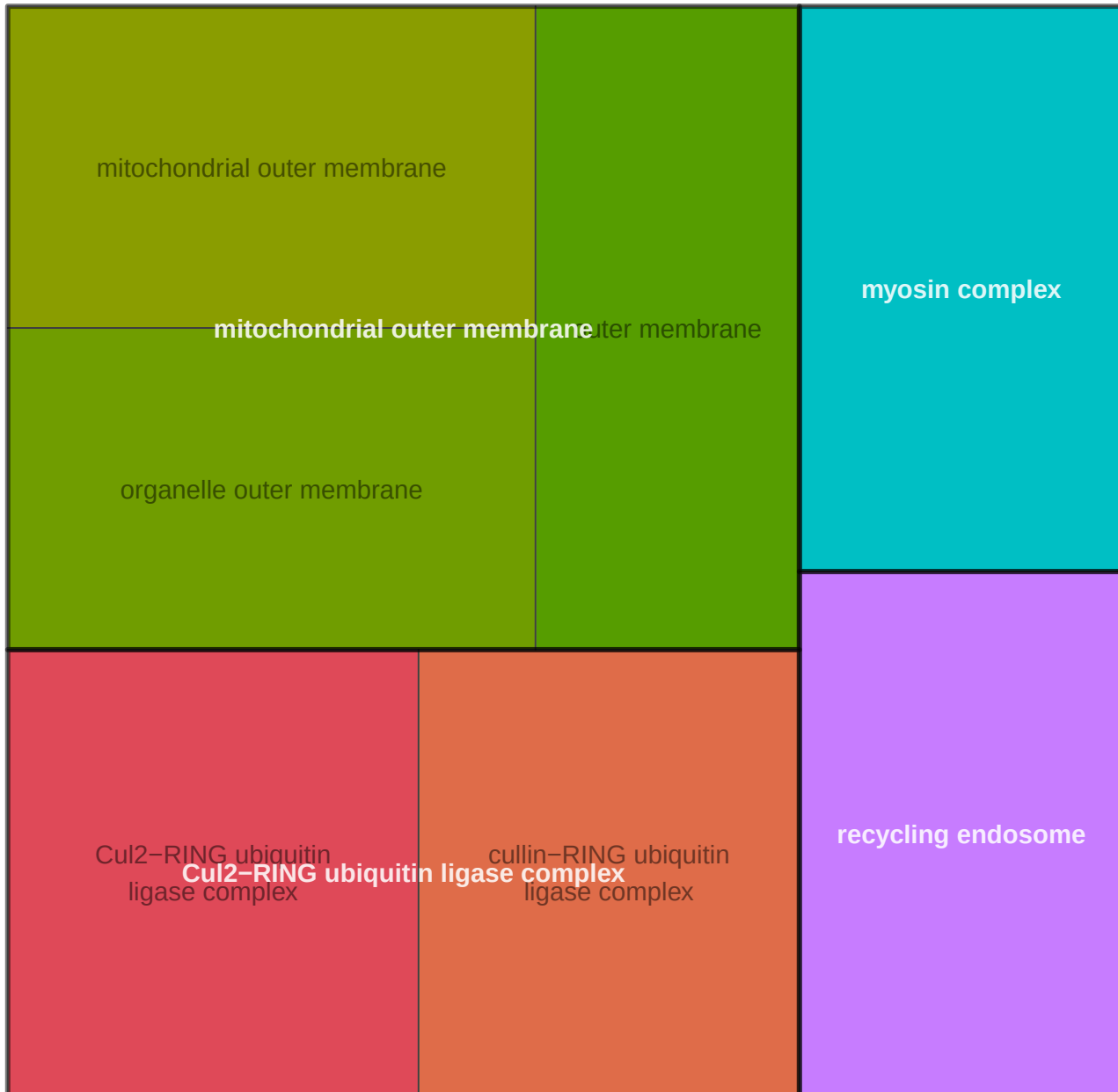
# Oligo-3\_M\_up BP



# Oligo-3\_M\_up MF



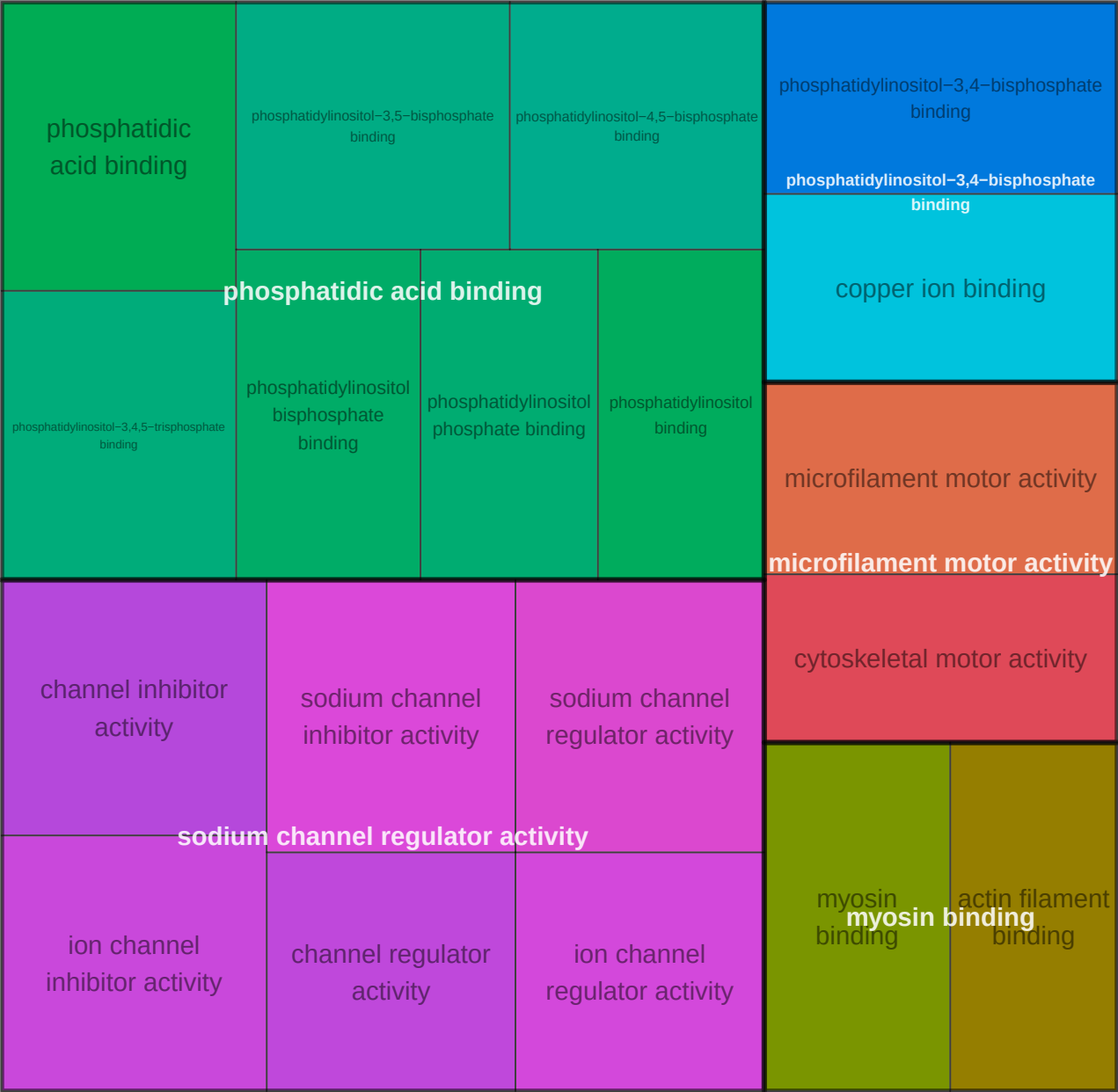
# Oligo-5\_0\_downF\_DownM CC



# Oligo-5\_0\_downF\_DownM BP



Oligo-5\_0\_downF\_DownM MF



Oligo-5\_0\_upF\_UpM MF

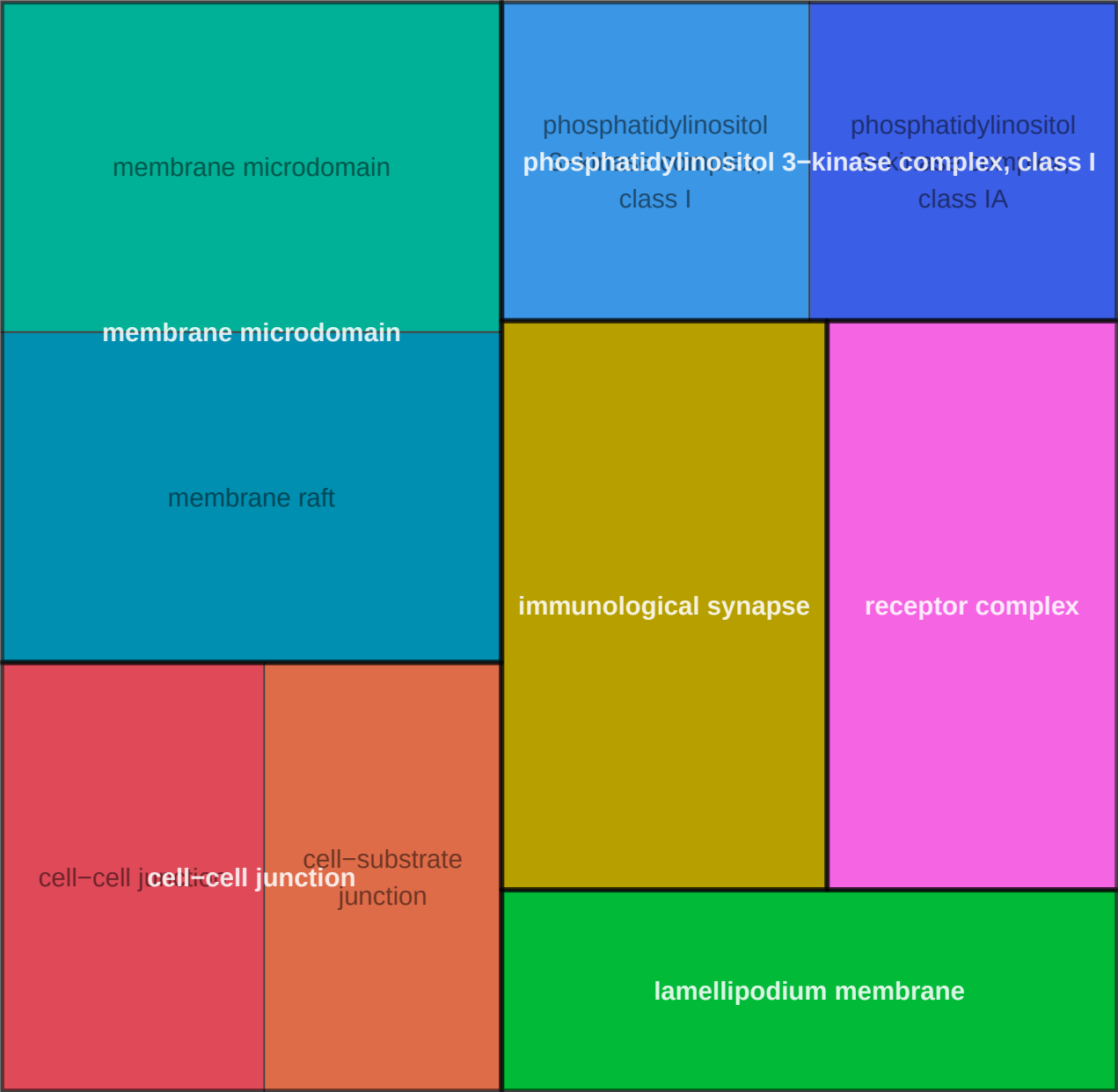
|                                                     |                                              |                                                        |                                                      |                                                         |                                                    |
|-----------------------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| adrenergic receptor activity                        | ephrin receptor activity                     | epidermal growth factor receptor activity              | fibroblast growth factor receptor activity           | G protein-coupled amine receptor activity               | neurotrophin binding                               |
| alpha-adrenergic receptor activity                  | GPI-linked ephrin receptor activity          | macrophage colony-stimulating factor receptor activity | neurotrophin receptor activity                       | placental growth factor receptor activity               |                                                    |
| boss receptor activity                              | hepatocyte growth factor receptor activity   | platelet-derived growth factor activity                | protein tyrosine kinase collagen receptor activity   |                                                         | S100 protein binding                               |
| brain-derived neurotrophic factor receptor activity | insulin receptor activity                    | platelet-derived growth factor beta-receptor activity  | transmembrane-ephrin receptor activity               | transmembrane receptor protein tyrosine kinase activity |                                                    |
| collagen receptor activity                          |                                              |                                                        |                                                      | protein-hormone receptor activity                       |                                                    |
|                                                     | insulin-like growth factor receptor activity | platelet-derived growth factor receptor activity       | vascular endothelial growth factor receptor activity | transmembrane receptor protein kinase activity          | structural molecule activity conferring elasticity |
| membrane scission                                   |                                              | membrane scission GTPase motor activity                |                                                      | molecular sequestering activity                         |                                                    |



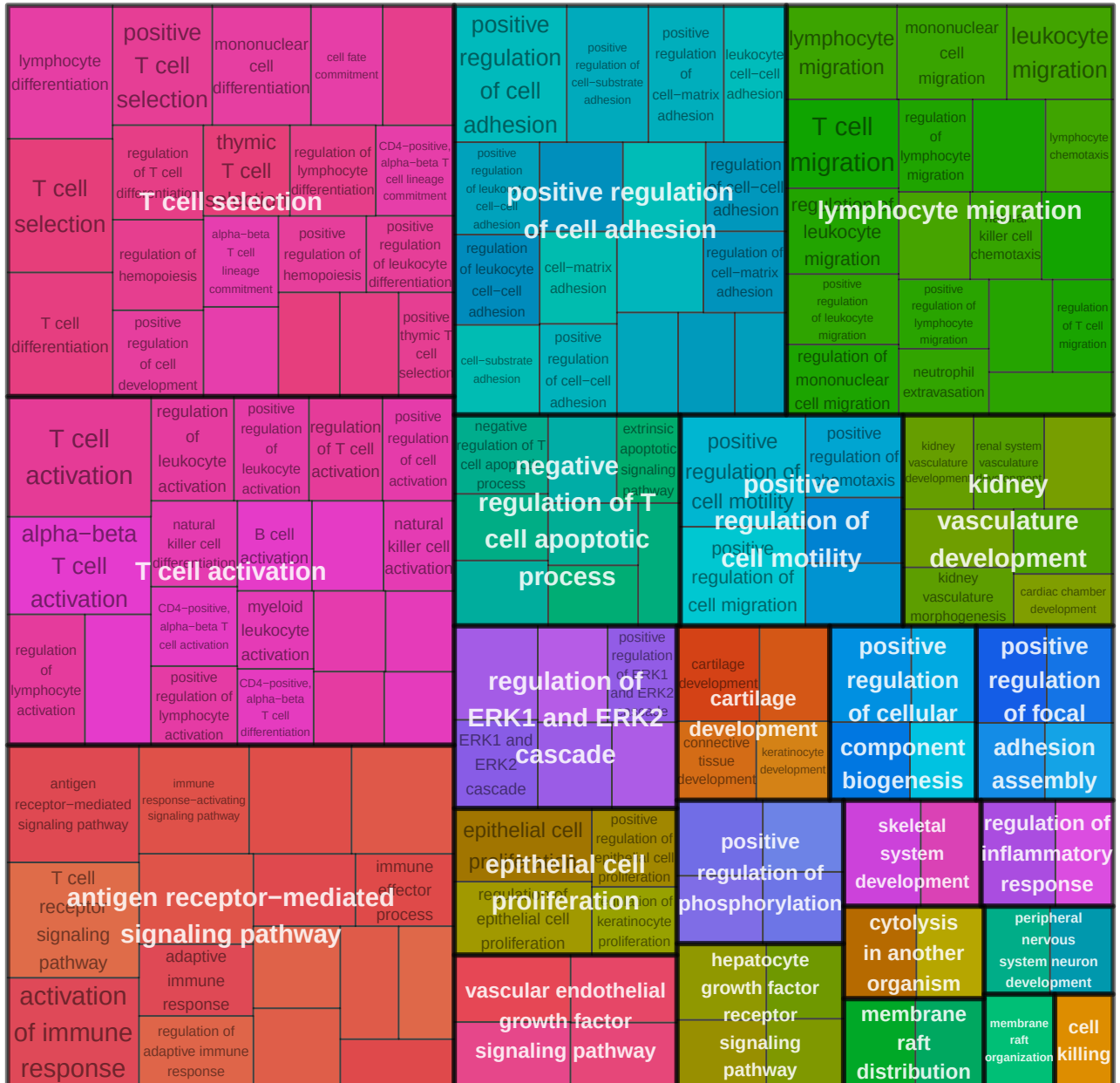
## Oligo-5\_M\_down CC

collagen type IX trimer collagen type IX trimer FACIT collagen trimer

Oligo-5\_M\_up CC



Oligo-5\_M\_up BP



## Oligo-5\_M\_up MF

protein kinase activity

transcription coactivator binding

transmembrane receptor protein  
protein kinase activity  
tyrosine kinase activity

transcription coactivator binding

transcription coregulator binding

vascular endothelial growth factor receptor activity

## OPC-5\_F\_up CC

anaphase-promoting complex

anaphase-promoting complex

nuclear ubiquitin ligase complex

cullin-RING ubiquitin  
ligase complex

OPC-5\_F\_up BP



Vasc-Endo\_0\_downF\_DownM BP



# Vasc-Endo\_0\_downF\_DownM MF

calcium, potassium:sodium  
antiporter activity

calcium:monoatomic cation  
antiporter activity

calcium:monoatomic cation antiporter activity

STAT family  
protein binding

calcium:sodium antiporter activity

solute:potassium antiporter activity



# Vasc-Endo\_0\_downF\_UpM CC



|                                                                            |                                          |                                                                                                 |                                        |                                                                                        |                                                                                                  |                                     |                                                        |                                           |
|----------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------|-------------------------------------------|
| regulation of relaxation of cardiac muscle                                 | regulation of relaxation of muscle       | negative regulation of blood circulation                                                        | interleukin-5 production               | positive regulation of interleukin-5 production                                        | cAMP catabolic process                                                                           | cyclic nucleotide catabolic process | vestibulocochlear nerve development                    | cranial nerve morphogenesis               |
| negative regulation of heart contraction                                   | relaxation of cardiac muscle             | positive regulation of heart contraction                                                        | regulation of interleukin-5 production | positive regulation of interleukin-2 production                                        | cAMP catabolic process                                                                           |                                     | vestibulocochlear nerve development                    |                                           |
| positive regulation of heart rate                                          | relaxation of muscle                     | cell communication involved in cardiac conduction                                               | interleukin-2 production               | positive regulation of type II interferon production                                   | cyclic nucleotide metabolic process                                                              | bile acid metabolic process         | inner ear receptor cell differentiation                | mechanoreceptor differentiation           |
| calcium ion transmembrane transport via high voltage-gated calcium channel | regulation of monoatomic anion transport | regulation of monoatomic ion transport                                                          | aminophospholipid translocation        | apical protein localization                                                            | negative regulation of adenylate cyclase-activating G protein-coupled receptor signaling pathway |                                     | regulation of plasma membrane organization             | phospholipid translocation                |
|                                                                            |                                          | release of sequestered calcium ion into cytosol by sarcoplasmic reticulum calcium ion transport | aminophospholipid transport            | bile acid and bile salt transport                                                      | response to epinephrine stimulus                                                                 | response to epinephrine stimulus    | regulation of lipid translocation                      | regulation of membrane lipid distribution |
| regulation of chloride transport                                           |                                          |                                                                                                 | microvillus assembly                   | regulation of microvillus organization                                                 | cellular response to epinephrine stimulus                                                        | response to epinephrine stimulus    | regulation of cardiac muscle cell contraction          | xenobiotic transmembrane transport        |
| regulation of chloride transport                                           |                                          |                                                                                                 | regulation of microvillus assembly     | microvillus organization                                                               | cellular response to cAMP                                                                        |                                     |                                                        |                                           |
| dense core granule exocytosis                                              | synaptic vesicle docking                 | dense core granule localization                                                                 |                                        | regulation of cell communication by electrical coupling involved in cardiac conduction | calcium ion-regulated exocytosis of neurotransmitter                                             |                                     | negative regulation of peptidyl-serine phosphorylation |                                           |
| neuronal dense core exocytosis                                             | secretory granule localization           |                                                                                                 |                                        |                                                                                        | calcium ion-regulated exocytosis of neurotransmitter                                             |                                     | establishment of endothelial barrier                   |                                           |
| presynaptic dense core vesicle exocytosis                                  | vesicle docking involved in exocytosis   | vesicle docking                                                                                 |                                        |                                                                                        |                                                                                                  |                                     |                                                        |                                           |

# Vasc-Endo\_0\_downF\_UpM MF



# Vasc-Endo\_0\_upF\_DownM CC



Vasc-Endo\_0\_upF\_DownM BP

|                                                       |                                         |                                               |                                           |                                                |
|-------------------------------------------------------|-----------------------------------------|-----------------------------------------------|-------------------------------------------|------------------------------------------------|
| protein maturation                                    | membrane protein ectodomain proteolysis | sensory perception of light stimulus          | sensory perception of mechanical stimulus | positive regulation of Notch signaling pathway |
| membrane protein proteolysis                          | protein maturation                      | sensory perception of sound                   |                                           | positive regulation of Notch signaling pathway |
| regulation of membrane protein ectodomain proteolysis | protein folding                         | sensory perception of sound                   | visual perception                         | regulation of Notch signaling pathway          |
| chloride transmembrane transport                      | chloride transport                      | peptidyl-histidine modification               |                                           | female pregnancy                               |
| inorganic anion transmembrane transport               | inorganic anion transport               | protein histidyl modification to diphthamide  |                                           | multi-organism reproductive process            |
| inorganic anion transport                             | monoatomic anion transport              | positive regulation of ATP-dependent activity | regulation of ATP-dependent activity      | multi-multicellular organism process           |

chloride channel activity

inorganic anion transmembrane transporter activity

monoatomic anion channel activity

ferrous iron binding

inorganic anion transmembrane transporter activity

chloride transmembrane transporter activity

monoatomic anion transmembrane transporter activity

ferrous iron binding

iron ion binding

ATPase activator activity

ATPase activator activity

ATPase activator activity

[illegible][illegible][illegible][illegible][illegible][illegible]

chloride channel activity

inorganic anion transmembrane transporter activity

monoatomic anion channel activity

ferrous iron binding

chloride transmembrane transporter activity

monoatomic anion transmembrane transporter activity

ferrous iron binding

iron ion binding

ATPase activator activity

ATPase activator activity

ATPase activator activity

[illegible]

chloride channel activity

inorganic anion transmembrane transporter activity

monoatomic anion channel activity

ferrous iron binding

chloride transmembrane transporter activity

monoatomic anion transmembrane transporter activity

ferrous iron binding

iron ion binding

ATPase activator activity

ATPase activator activity

ATPase activator activity

chloride channel activity

inorganic anion transmembrane transporter activity

monoatomic anion channel activity

ferrous iron binding

chloride transmembrane transporter activity

monoatomic anion transmembrane transporter activity

ferrous iron binding

iron ion binding

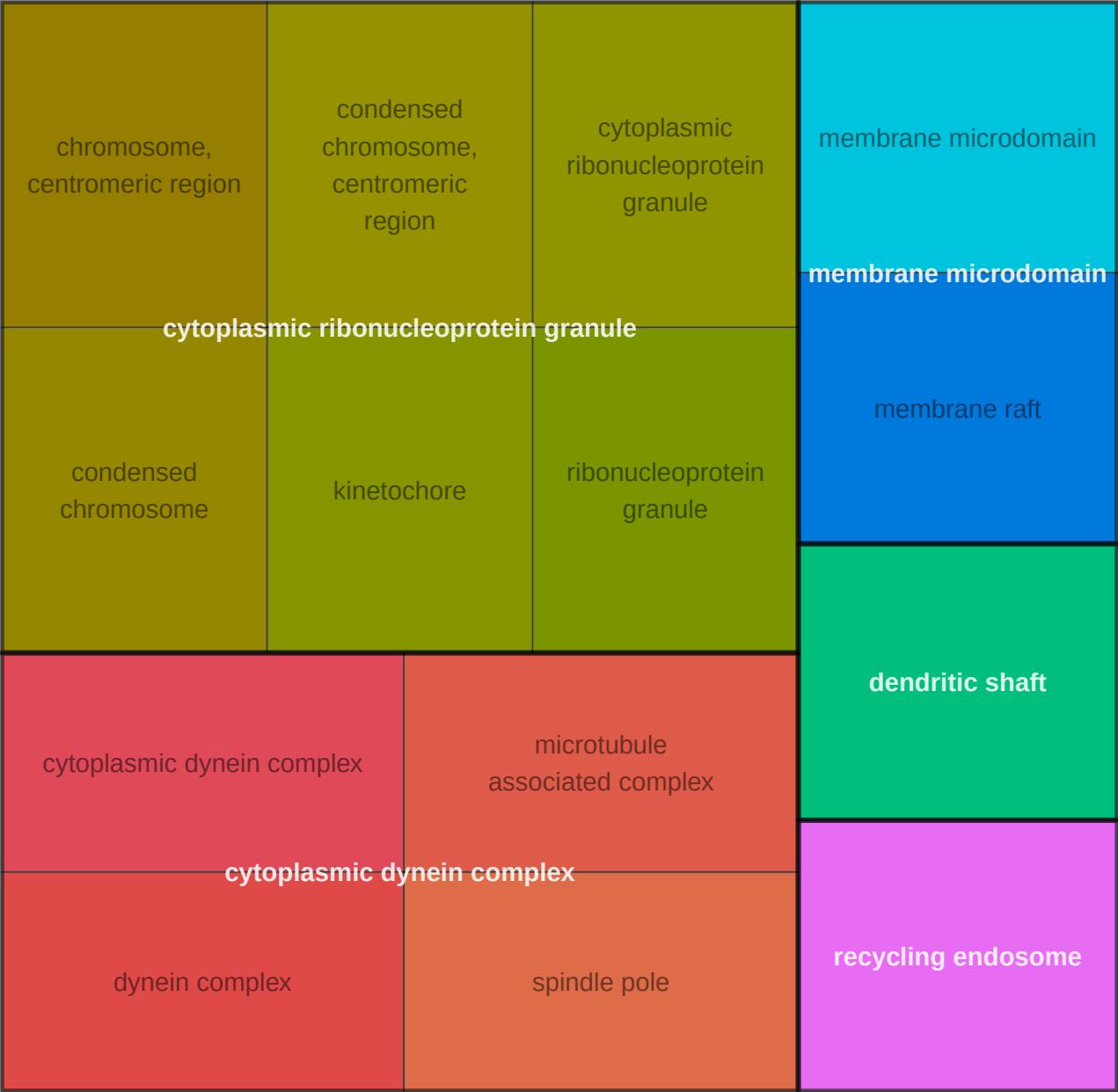
ATPase activator activity

ATPase activator activity

ATPase activator activity

[illegible][illegible]

Vasc-Endo\_0\_upF\_UpM CC

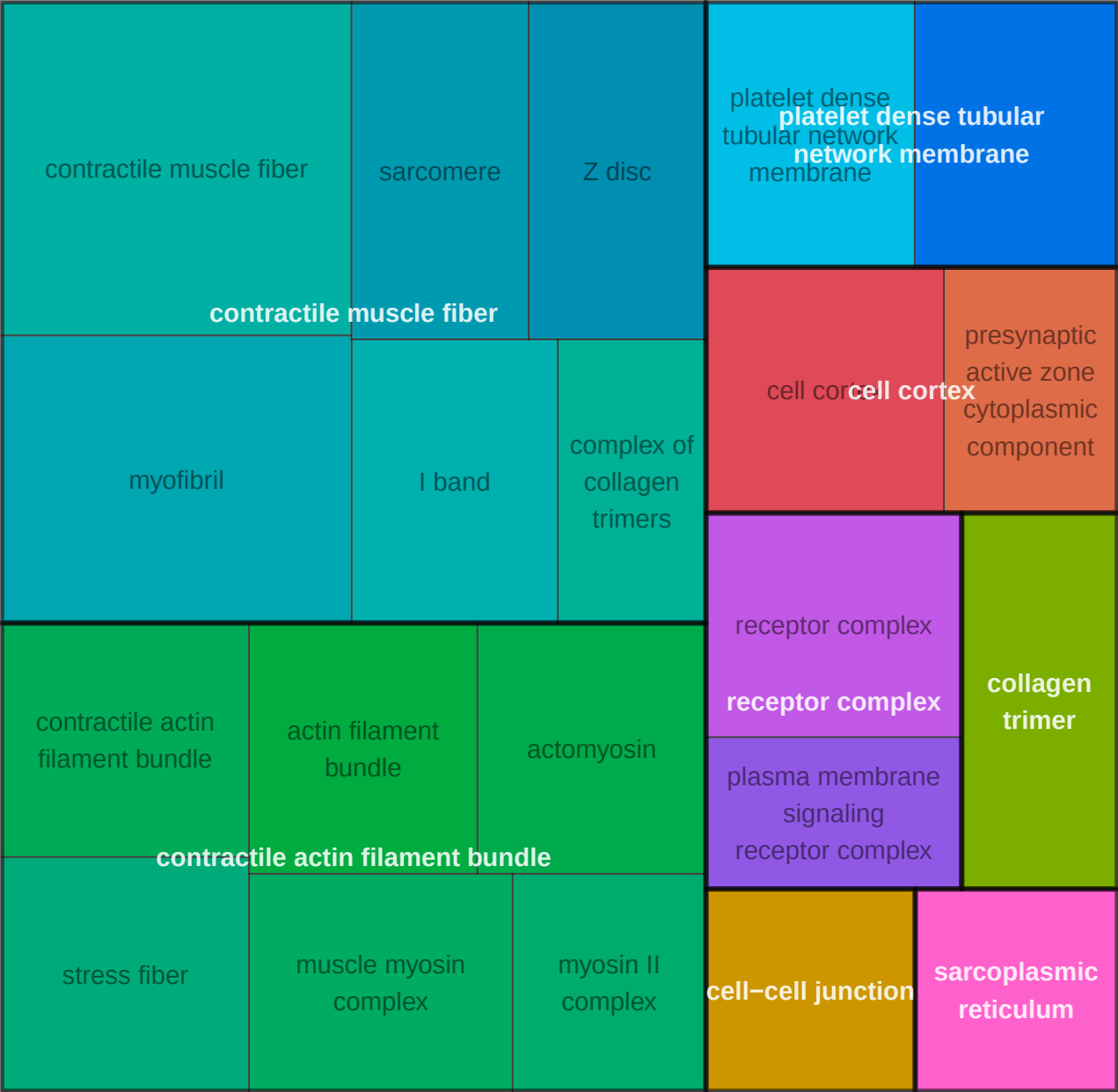


Vasc-Endo\_0\_upF\_UpM BP





Vasc-Endo\_M\_up CC



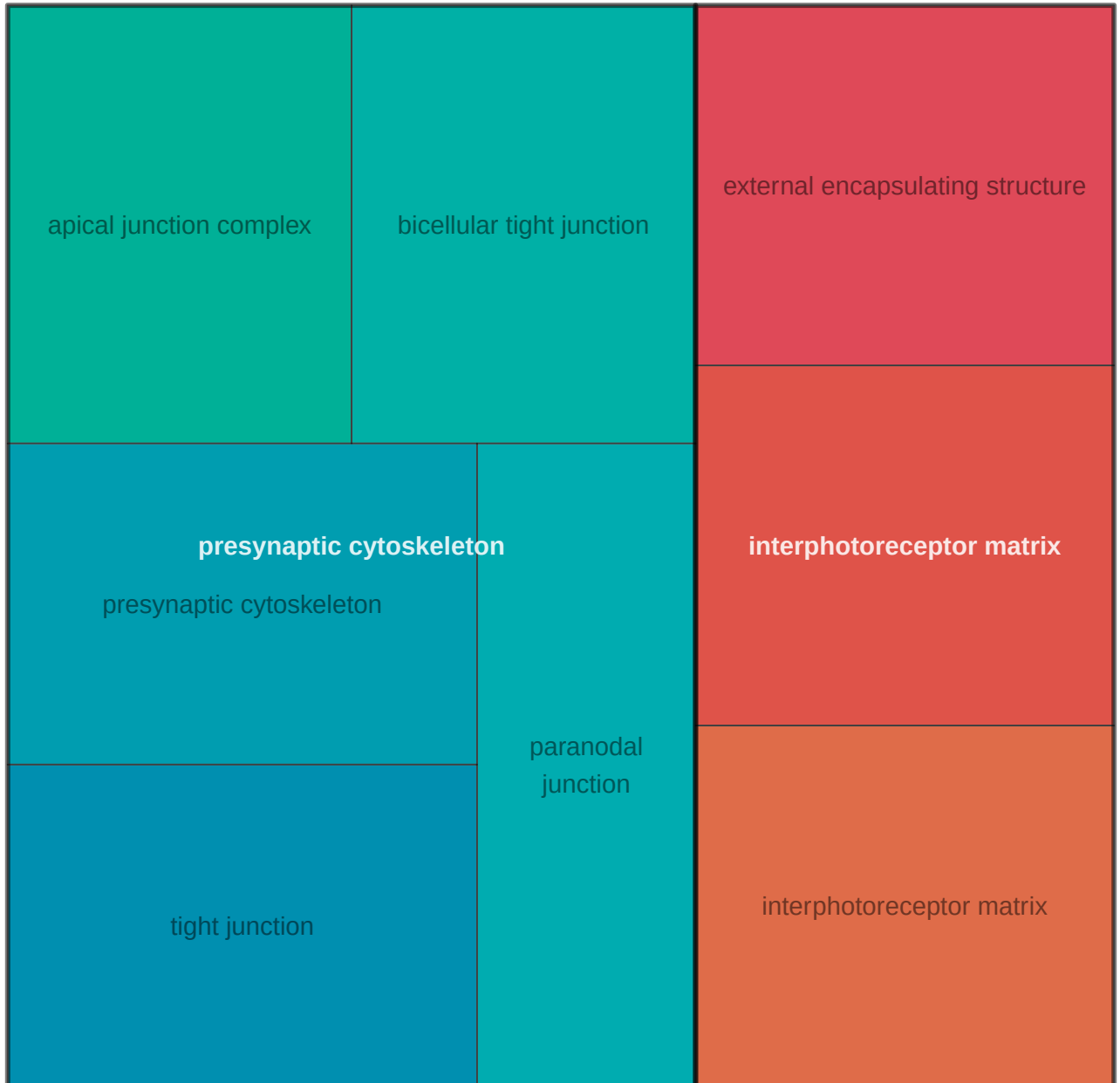
Vasc-Endo\_M\_up BP



Vasc-Endo\_M\_up MF



# Vasc-VLMC\_0\_downF\_UpM CC



# Vasc-VLMC\_0\_downF\_UpM MF

ciliary neurotrophic  
factor receptor binding

neuroligin family  
protein binding

**calcium-activated  
potassium channel activity**

**ciliary neurotrophic factor receptor binding**

integrin binding

Roundabout binding

**profilin binding**

**laminin-1 binding**

**structural constituent of  
presynaptic active zone**

**GTPase inhibitor activity**

# Vasc-VLMC\_M\_up CC

external encapsulating structure

extracellular matrix

**external encapsulating structure**

**desmosome**

collagen-containing extracellular matrix